

Discrete Distributions: Measure Foundations to Exam P

Reusable seven-session packet – 810 minutes – ENS L3 end / early M1 discipline

This cycle is independent of the July measure-theory schedule. Its mathematical order is fixed:

atomic measure $\longrightarrow$ fibre pushforward $\longrightarrow$ joint law and convolution
 $\longrightarrow$ named constructions $\longrightarrow$ restriction and mixture $\longrightarrow$ Exam P deployment.

The family name is the conclusion of a construction, not the first line of a calculation.

Contract

- **Objects:** countable laws, counting-measure densities, joint atomic laws, convolution, sampling constructions, waiting laws, Poisson mechanisms, restriction, and mixtures.
- **Purpose:** recover every Exam P discrete calculation from a legitimate measure, pushforward, product construction, or normalized restriction.
- **Regime:** ENS theorem boundaries and pathological counterexamples, with Exam P used as the terminal speed test.
- **Required output:** seven written attempts, two ledgers, a first rupture and minimal repair, one oral examination, and one sixty-minute official audit.
- **Boundary:** no generating functions or MGFs; the CLT appears only as an endpoint approximation.

Work schedule

Day	Time	Assigned work	Closed-book responsibility
J1	90m	15m entry gate; 30m Cours 1–3; 35m TD 0–1; 10m ledger	Atomic representation, discrete LOTUS, and the uncountable counting-measure failure
J2	105m	20m Cours 4–5; 60m TD 2–3; 15m Exercise A; 10m record	Fibre law, addition pushforward, and convolution only under the product joint law
J3	120m	25m Cours 6.1, 6.4, 6.5; 70m TD 4; 15m Exercise B; 10m record	Binomial and hypergeometric constructions; covariance origin of the finite-population factor
J4	105m	20m Cours 6.3; 60m TD 5; 15m Exercise C; 10m record	Waiting-time/count duality, convention shift, and finite-population failure
J5	120m	25m Cours 6.2; 70m TD 6; 15m Exercise D; 10m record	Poisson factorial moments, superposition, splitting, and conditional allocation
J6	120m	20m Cours 7; 65m TD 7; 20m Exercise E; 15m ledger	Mixture variance, normalized restriction, truncation, and Bayes reversal
J7	150m	45m TD 8; 30m oral; 60m official audit; 15m verdict	Latent class to thinned mixture to nonlinear payment, compressed to exam speed

Official audit

Use the official SOA sample-question document without reproducing its statements. Lock all eight answers before opening the official solutions.

Question	Limit	Structural object
Q30	7m	fixed independent Bernoulli trials
Q128	7m	hypergeometric count
Q146	8m	finite-horizon negative-binomial event and conditioning
Q150	8m	discrete-uniform conditional mixture
Q227	8m	Poisson mixture variance
Q246	8m	independent Poisson components conditioned on a total
Q386	7m	nonlinear Poisson factorial moment
Q512	7m	second moment versus squared mean

Question numbers are snapshot anchors. Q149, Q243, Q314, and Q382 are mastery extensions and do not belong to the 810-minute core.

Solution discipline

The canonical TD corrigé remains closed until the corresponding written attempt is complete. The independent solution PDF answers only the packet gates. Locate the first illegitimate step, close the solution, and reconstruct that step without notes on the following day.

Deployable verdict

The cycle is deployable only if all five conditions hold:

1. the atomic representation, convolution boundary, hypergeometric variance, and Poisson splitting are recovered without notes;
2. binomial, geometric, negative binomial, hypergeometric, Poisson, and finite uniform laws are constructed before being named;
3. every pushforward, sum, conditioning, mixture, and truncation is located on the correct measure;
4. the official audit score is at least seven out of eight, with no structural model error;
5. one oral sujet and at least two boundary relances are completed within the stated time.

Selected reconstruction notes

The theorem statements required by the weekly packet; complete proofs remain in the canonical Cours and TD

1. Atomic representation

Let E be finite or countably infinite with counting measure $\#_E$. For a probability measure μ on $(E, 2^E)$, set

$$p(x) = \mu(\{x\}).$$

Then

$$\mu(A) = \sum_{x \in A} p(x), \quad \mu = p\#_E, \quad \sum_{x \in E} p(x) = 1.$$

The density is unique $\#_E$ -almost everywhere and therefore pointwise. For every nonnegative φ ,

$$\int_E \varphi d\mu = \sum_{x \in E} \varphi(x)p(x).$$

For signed φ , finiteness is equivalent to $\sum_E |\varphi|p < \infty$.

Boundary. On an uncountable space, full counting measure is not σ -finite. Every measure is absolutely continuous with respect to it, but a nonatomic probability cannot equal $f\#$. Absolute continuity outside the RN theorem regime is not a density theorem.

2. Pushforwards and fibres

If X has law μ_X on a countable set E and $Y = g(X)$, then

$$\mu_Y = g\#\mu_X, \quad p_Y(y) = \sum_{x \in g^{-1}(\{y\})} p_X(x).$$

For every admissible payment g ,

$$\mathbb{E}[g(X)] = \int_E g d\mu_X = \sum_{x \in E} g(x)p_X(x).$$

The identity does not imply $\mathbb{E}[g(X)] = g(\mathbb{E}[X])$. A non-injective pushforward records only the total mass of each fibre and destroys the allocation inside the fibre.

3. Joint laws and convolution

For countable-valued (X, Y) ,

$$p_X(x) = \sum_y p_{X,Y}(x, y), \quad p_Y(y) = \sum_x p_{X,Y}(x, y).$$

Independence is the measure equality

$$\mu_{X,Y} = \mu_X \otimes \mu_Y,$$

equivalently

$$p_{X,Y}(x, y) = p_X(x)p_Y(y).$$

For integer-valued variables, the addition map gives

$$p_{X+Y}(n) = \sum_{k \in \mathbb{Z}} p_{X,Y}(k, n-k).$$

Only under independence does this become

$$p_{X+Y} = p_X * p_Y.$$

Identical marginals can support different couplings and hence different sum laws.

If $p_X(x) > 0$, the conditional fibre law is

$$p_{Y|X=x}(y) = \frac{p_{X,Y}(x, y)}{p_X(x)}.$$

The numerator is a restricted joint measure and the denominator normalizes its mass. If $p_X(x) = 0$, this elementary ratio is undefined.

4. Canonical families as constructions

Write $q = 1 - p$.

Bernoulli and binomial

If $B_1, \dots, B_n$ are iid Bernoulli(p) and $S_n = \sum_i B_i$, then

$$\mathbb{P}(S_n = k) = \binom{n}{k} p^k q^{n-k}, \quad 0 \leq k \leq n,$$

and

$$\mathbb{E}[S_n] = np, \quad \text{Var}(S_n) = npq.$$

Independence and a common success parameter are both part of the construction. Independent Bernoulli variables with unequal parameters generally produce a Poisson-binomial law.

Geometric and negative binomial

For iid Bernoulli(p) trials with $0 < p \leq 1$,

$$G = \min\{n \geq 1 : B_n = 1\}$$

has

$$\mathbb{P}(G = k) = q^{k-1}p, \quad \mathbb{E}[G] = \frac{1}{p}, \quad \text{Var}(G) = \frac{q}{p^2}.$$

The trial of the r -th success,

$$T_r = \min \left\{ n \geq r : \sum_{j=1}^n B_j = r \right\},$$

satisfies

$$\mathbb{P}(T_r = n) = \binom{n-1}{r-1} p^r q^{n-r}, \quad n \geq r,$$

with

$$\mathbb{E}[T_r] = \frac{r}{p}, \quad \text{Var}(T_r) = \frac{rq}{p^2}.$$

Moreover,

$$\{T_r \leq n\} = \left\{ \sum_{j=1}^n B_j \geq r \right\}.$$

The failures-before-success convention is $T_r - r$ and has support $\mathbb{N}$. If $p = 0$, the hitting time is infinite; if $p = 1$, $T_r = r$ almost surely.

Hypergeometric

From a population of size N with K successes, draw a uniform sample of size n without replacement. The number H of sampled successes has

$$\mathbb{P}(H = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$$

on

$$\max\{0, n - N + K\} \leq k \leq \min\{n, K\}.$$

For $N > 1$,

$$\mathbb{E}[H] = n \frac{K}{N}, \quad \text{Var}(H) = n \frac{K}{N} \left(1 - \frac{K}{N} \right) \frac{N-n}{N-1}.$$

The last factor is generated by negative off-diagonal covariances. When $N = 1$, the construction is deterministic and the displayed quotient is not used.

The position $T_r^{(N)}$ of the r -th success in a finite sequence with exactly K successes has

$$\mathbb{P}(T_r^{(N)} = t) = \frac{\binom{t-1}{r-1} \binom{N-t}{K-r}}{\binom{N}{K}}, \quad r \leq t \leq N - K + r.$$

Its bounded support and dependent draws exclude the negative-binomial construction.

Poisson

For $N \sim \text{Pois}(\lambda)$,

$$\mathbb{P}(N = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k \in \mathbb{N},$$

with $\text{Pois}(0) = \delta_0$. Index shifts give

$$\mathbb{E}[N] = \lambda, \quad \mathbb{E}[(N)_2] = \lambda^2, \quad \text{Var}(N) = \lambda.$$

If N_1, N_2 are independent Poisson variables, then

$$N_1 + N_2 \sim \text{Pois}(\lambda_1 + \lambda_2).$$

Independent Bernoulli labels applied to a Poisson count produce independent counts with parameters λp and $\lambda(1-p)$. Conversely, if $\lambda_1 + \lambda_2 > 0$,

$$N_1 \mid (N_1 + N_2 = n) \sim \text{Bin}\left(n, \frac{\lambda_1}{\lambda_1 + \lambda_2}\right).$$

Every statement has an independence hypothesis at the point where a joint mass factorizes.

Finite uniform law

For a nonempty finite set A ,

$$\text{Unif}(A) = \frac{1}{|A|} \#_A.$$

No countably infinite set supports a probability law assigning one common mass to every point.

5. Restriction, mixture, and truncation

For $A \subseteq E$ with $\sum_{x \in A} p(x) > 0$,

$$p_A(x) = \frac{p(x) \mathbf{1}_A(x)}{\sum_{z \in A} p(z)}.$$

This is restriction followed by normalization. If the denominator vanishes, the elementary conditional law is undefined.

For a latent state I with weights α_i and conditional pmfs p_i ,

$$p_X(x) = \sum_i \alpha_i p_i(x).$$

If $X \in L^2$ and the conditional means and variances are m_i, v_i , then

$$\text{Var}(X) = \sum_i \alpha_i v_i + \sum_i \alpha_i (m_i - \mathbb{E}[X])^2.$$

In particular, if $N \mid \Lambda \sim \text{Pois}(\Lambda)$ and $\Lambda \in L^2$,

$$\text{Var}(N) = \mathbb{E}[\Lambda] + \text{Var}(\Lambda).$$

A non-degenerate square-integrable mixing intensity therefore produces overdispersion and cannot yield one Poisson law.

Zero truncation is a different operation. For $M \sim \text{Pois}(\lambda)$ with $\lambda > 0$,

$$\mathbb{P}(M = k \mid M \geq 1) = \frac{e^{-\lambda} \lambda^k}{(1 - e^{-\lambda}) k!}, \quad k \geq 1.$$

The support and normalization change, so the result is not a Poisson law.

6. Nonlinear moments and terminal approximation

For an integer-valued count,

$$N^2 = (N)_2 + N.$$

Thus, for a Poisson variable,

$$\mathbb{E}\left[\frac{N(N+1)}{2}\right] = \frac{\lambda^2}{2} + \lambda.$$

Evaluating the polynomial at $\mathbb{E}[N]$ deletes half the variance.

For iid X_i with mean m and variance $0 < \sigma^2 < \infty$,

$$\frac{\sum_{i=1}^n X_i - nm}{\sigma\sqrt{n}} \Rightarrow \mathcal{N}(0, 1).$$

For a unit-lattice sum, the continuity correction belongs to the approximation map. The CLT is not an exact pushforward identity and supplies no universal finite- n error bound.

7. Minimal reconstruction order

Before arithmetic, expose

state space $\rightarrow$ law construction $\rightarrow$ support $\rightarrow$ pushforward or restriction $\rightarrow$ finite sum.

Independence, replacement, and positive-denominator hypotheses must appear where they act. Naming the family after the calculation is too late.

Seven-Day Problem Sheet — Discrete Laws and Structural Deployment

The canonical TD carries the substantial proof responsibility. The exercises below are short transfer gates: they test whether the objects constructed in the TD remain available when the notation, numerical data or verbal model changes. Complete each gate without the canonical corrigé.

Canonical assignment map

Day	Canonical written work	Packet gate
J1	TD 03, Exercises 0–1	Entry Gate
J2	TD 03, Exercises 2–3	Exercise A
J3	TD 03, Exercise 4	Exercise B
J4	TD 03, Exercise 5	Exercise C
J5	TD 03, Exercise 6	Exercise D
J6	TD 03, Exercise 7	Exercise E
J7	TD 03, Exercise 8; one Colle sujet	Official Audit and Reconstruction

Entry Gate — Fifteen-minute recognition test

No proof longer than four lines is requested in this gate.

- Let E be countable and let μ be a probability measure on $(E, 2^E)$. Define the function p which represents μ with respect to counting measure, state the representation of $\mu(A)$, and give the exact uniqueness basis.
- Let $\#$ be counting measure on the uncountable Borel space $[0, 1]$. State why Lebesgue measure λ satisfies $\lambda \ll \#$ and why no measurable f can satisfy $\lambda = f\#$.
- Let $Y = g(X)$. Write the legitimate formula for $\mathbb{E}[Y]$ in terms of the law of X , and state the invalid interchange which it does not imply.
- Let X, Y be integer-valued. Write the law of $X + Y$ first in terms of the joint pmf, then state the additional hypothesis which turns the formula into convolution of the marginals.
- Match each verbal construction with a law or class of laws:
 - a fixed number of iid Bernoulli trials;
 - a uniform sample without replacement from a finite population;
 - the trial on which the r -th success occurs;
 - a Poisson count with a non-degenerate square-integrable random intensity.

If one of the four objects—density, pushforward, joint law or construction—cannot be named, record it before reading.

Exercise A — Identical marginals, different pushforwards

Let $B \sim \text{Bernoulli}(1/2)$. Define

$$(X_+, Y_+) := (B, B), \quad (X_-, Y_-) := (B, 1 - B).$$

- Prove that all four one-dimensional marginals are Bernoulli(1/2).
- Determine the joint pmf and the sum law in both models.
- Determine which, if either, of the two pairs is independent.

- Let $h(x, y) = \mathbf{1}_{\{x+y \geq 1\}}$. Compute the law and expectation of $h(X_+, Y_+)$ and $h(X_-, Y_-)$. Identify the measure being pushed forward in each calculation.
- State the strongest conclusion about $X + Y$ which can be obtained from Bernoulli(1/2) marginals alone.

Limit: 15 minutes after the canonical TD attempt.

Exercise B — Finite-population correction in a numerical model

A population contains $N = 20$ objects, of which $K = 7$ are successes. A sample of size $n = 5$ is drawn uniformly without replacement, and H denotes the number of sampled successes.

- State the exact support and pmf of H .
- Construct H from ordered-draw indicators and compute $\mathbb{E}[H]$.
- Compute $\text{Var}(H)$, first in structural form and then as an exact rational number.
- Compare this variance with that of a binomial(5, 7/20) variable. Identify the covariance term which accounts for the difference.
- State what happens to the variance formula if $n = N$, and separately explain why the displayed quotient is not used when $N = 1$.

Limit: 15 minutes after the canonical TD attempt.

Exercise C — Finite and infinite waiting structures

A finite sequence of $N = 12$ positions contains exactly $K = 5$ successes, and the success set is uniform among the 5-subsets of $\{1, \dots, 12\}$. Let $T_2^{(12)}$ be the position of the second success.

- Determine the exact support of $T_2^{(12)}$.
- Derive its pmf from the number of admissible success sets.
- Compare it with the trial-convention negative-binomial variable T_2 generated by iid Bernoulli(5/12) trials. Give two structural reasons why the laws are different.
- For the iid model, prove

$$\{T_2 \leq n\} = \{\text{Bin}(n, 5/12) \geq 2\}.$$

- Convert T_2 to the failures-before-second-success convention and state the new support. Treat the endpoints $p = 0$ and $p = 1$ separately.

Limit: 15 minutes after the canonical TD attempt.

Exercise D — One Poisson model through three representations

Let $N \sim \text{Pois}(4)$. Conditional on $N = n$, independently label every event A with probability 1/3 and B with probability 2/3. Let K and L be the type counts.

- Derive the joint pmf of (K, L) and factorize it.
- Identify the marginal laws and prove that K and L are independent.
- Determine the conditional law of K given $K + L = 5$.
- For

$$Y = \frac{K(K+1)}{2},$$

compute $\mathbb{E}[Y]$ using factorial moments.

- Compute $g(\mathbb{E}[K])$ for $g(k) = k(k+1)/2$ and identify the missing term.
- State exactly which conclusions fail if the type labels are no longer conditionally independent.

Limit: 15 minutes after the canonical TD attempt.

Exercise E — Mixture and truncation are different operations

Let $I \sim \text{Bernoulli}(1/4)$ and, conditionally on $I = i$, let

$$N \mid I = 0 \sim \text{Pois}(1), \quad N \mid I = 1 \sim \text{Pois}(4).$$

- Derive the unconditional pmf of N .

2. Compute $\mathbb{E}[N]$ and $\text{Var}(N)$ without summing the mixture pmf term by term.
3. Prove that the unconditional law is not Poisson.
4. Compute $\mathbb{P}(I = 1 \mid N = 0)$, naming the restricted masses before normalizing.
5. Independently, let $M \sim \text{Pois}(\lambda)$ with $\lambda > 0$. Derive the pmf, mean and variance of M conditional on $M \geq 1$.
6. Explain why the mixture in Questions 1–4 and the truncation in Question 5 both retain Poisson factors in their numerators but are produced by different operations on measures.

Limit: 20 minutes after the canonical TD attempt.

Official Audit — Sixty minutes

Use the official SOA sample-question PDF. Do not reproduce the statements in this vault. Before calculating, complete the first three columns; after locking an answer, complete the last two.

Question	State space and construction	Required pushforward, restriction or moment	Answer	First rupture, if any
Q30				
Q128				
Q146				
Q150				
Q227				
Q246				
Q386				
Q512				

Time limits are respectively 7, 7, 8, 8, 8, 8, 7, 7 minutes. Do not open the official solutions until all eight answers are locked.

Colle record

Choose one sujet from Colle 01. Use fifteen minutes for preparation and fifteen minutes for exposition.

- Sujet selected:
- Theorem or construction stated:
- First relance:
- Second relance:
- First point at which the exposition ceased to be reconstructible:

Closed-book reconstruction — Sixty minutes

This reconstruction is a mastery extension and may be completed after the 810-minute core.

1. Prove the atomic representation theorem and state the uncountable counting-measure counterexample.
2. Derive the fibre formula and the law of a sum as pushforwards.
3. Give identical marginals with different sum laws and locate the missing product-law hypothesis.
4. Construct binomial and hypergeometric laws from their probability spaces; derive the finite-population correction.
5. Derive geometric and negative-binomial waiting laws and state both conventions.
6. Derive Poisson factorial moments, superposition, splitting and conditional allocation without an MGF.
7. Separate normalized restriction, mixture and posterior reversal.
8. Complete the integrated chain

latent class $\rightarrow$ conditional count $\rightarrow$ thinning $\rightarrow$ mixture $\rightarrow$ nonlinear payment.

Source calibration

The packet gates are original. Their terminal mechanisms are calibrated against the July 2026 Exam P syllabus and the official SOA sample-question bank. Official problem statements are not reproduced. The source determines scope and terminal speed; the canonical Cours and TD determine proof responsibility.

Work and rupture record

Record the first illegitimate step, not a general impression of difficulty

Session record

Day	Planned	Actual	State	First rupture and minimal repair
J1	90m		not attempted	
J2	105m		not attempted	
J3	120m		not attempted	
J4	105m		not attempted	
J5	120m		not attempted	
J6	120m		not attempted	
J7	150m		not attempted	

Allowed states are: rupture, reconstructible, deployable. “Difficult” is not a diagnosis.

Atomic and pushforward interfaces

Counting measure, domain, and sigma-finite cover:	
Atomic density and uniqueness basis:	
Uncountable RN failure and missing hypothesis:	
Fibre map and source law:	
Addition map and source joint law:	
Product-law equality used for convolution:	
Same marginals, different sums counterexample:	

Replacement and waiting interfaces

With-replacement probability space:	
Without-replacement probability space:	
Exact hypergeometric support:	
Off-diagonal covariance and finite-population factor:	
Trial-convention waiting support:	
Failures-convention support:	
Finite waiting support and failure of negative-binomial construction:	
Endpoint $p = 0$ and endpoint $p = 1$:	

Family construction ledger

Family	Construction	Support	Mean / variance	Boundary or convention
Binomial				
Geometric				
Negative binomial				
Hypergeometric				
Poisson				
Finite uniform				

Poisson, restriction, and mixture interfaces

Index shift for $\mathbb{E}[N]$:	
Index shift for $\mathbb{E}[(N)_2]$:	
Joint factorization for splitting:	
Conditional allocation given a total:	
Normalized restriction numerator and denominator:	
Mixture component measures and weights:	

Within-component variance:
 Between-component variance:
 Zero-truncation normalizer:
 Posterior numerator masses:

Operation ledger

Task	Source measure	Operation and hypothesis	Output
Transformed payment			
Sum law			
Convolution			
Sample without replacement			
Positive-mass conditioning			
Latent mixture			
Nonlinear moment			
CLT approximation			

Core proof audit

Proof	Closed book	First missing object or theorem	Repaired next day
Atomic representation	yes / no		yes / no
Convolution boundary	yes / no		yes / no
Hypergeometric variance	yes / no		yes / no
Poisson splitting	yes / no		yes / no

Official audit

Q	Limit	Locked	Correct	Structural object	Error class
30	7m				
128	7m				
146	8m				
150	8m				
227	8m				
246	8m				
386	7m				
512	7m				

Score out of eight:
 Structural errors present: yes / no
 Official solutions opened only after all answers were locked: yes / no

Oral and final verdict

Sujet selected: A / B / C / D / E
 First relance and answer:
 Second relance and answer:
 Cycle state: rupture / reconstructible / deployable
 Four core proofs recovered without notes: yes / no
 Six required families constructed before being named: yes / no
 Operations located on the correct measures: yes / no
 Official audit at least seven out of eight: yes / no
 First remaining non-deferred rupture:
 Minimal valid repair: